

HYBRID NETWORK PROPOSAL

Own the network your lodge runs on.

A carrier-grade Vodacom link, matched to a carrier-grade private network across the reserve — engineered for Barefoot Addo Elephant Lodge to turn a rising monthly rental into infrastructure it owns, on its balance sheet, for the long term.

Prepared for

Barefoot Addo Elephant Lodge (Pty) Ltd

Prepared by

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Date

28 September 2026

01 Where things stand today

The lodge currently pays R18,105.60 a month for a shared, best-effort connection and a set of rented Wi-Fi access points. Nothing about that setup belongs to the lodge, and none of it scales without another monthly line item.

WHAT WE FOUND ON SITE

The backhaul radios that carry the whole property's connectivity are mounted on a leaning gum pole — not an engineered structure. It has held so far. It is not something to plan a lodge's connectivity around.

WHERE THE CURRENT NETWORK FALLS SHORT

The spa has no connectivity at all — it was never part of the original rollout. The furthest buildings, the staff village and the Director's House, sit 2.3–3 km out on the edge of a single 90° sector, which is where a rented network is weakest and least accountable. And every fix on the current model is the same answer: another rented access point, another monthly line item, on a shared connection that was never engineered for this property.

What we're building instead

A private wireless network on an engineered 9m mast, replacing the gum pole entirely. Two sector radios cover the full property from that single mast — the main lodge cluster at close range, and a second sector reaching out to the villas, the staff village, and the Director's House.

Twenty-eight access points go in across every building that needs coverage. That includes the spa — connected for the first time — alongside the boma, the school, the workshop, and every guest and staff area on the property.

The order of work matters. The private network — mast, cabinet, power, radios and access points — is built first. The Vodacom link is the final piece, and it can only go in once the mast is standing: Vodacom surveys the site against that mast before they will install.

BUILT FOR WHAT COMES NEXT

Because the sector coverage already reaches every corner of the property, adding CCTV later doesn't mean building new wireless infrastructure — it means plugging cameras into a network that's already live there. Security cameras at the boma, the villas, or the staff village become an endpoint cost, not an infrastructure project.

NOT A LOCAL FIX

CTTX is based in Port Elizabeth, but this build runs on Vodacom's national carrier network — over 14,000 towers across South Africa. If there's a second property, or a third, anywhere in the country, CTTX delivers and supports the same carrier-grade design there too. This isn't a one-off local install; it's a model that travels with the properties you own.

Why this has to work every time

A lodge network carries three different loads at once. This design was built to hold all three, not just the one that's easiest to sell.

CLIENT EXPERIENCE

Many guests are international, checking in with family across several time zones — sharing the day's sighting, confirming they've arrived safely, staying reachable somewhere their own cellphone has no signal. That only works if the connection is as reliable at 2am as it is at 2pm.

OPERATIONAL REQUIREMENT

Bookings, POS, staff communication and day-to-day lodge management all run over this same network. A dropped connection on a busy weekend isn't an inconvenience — it's a stalled payment or a missed booking.

SECURITY REQUIREMENT

Reserves deal with real incidents — a boundary breach, a security concern that surfaces after the office has closed for the night. When that happens, communication is what determines how fast a response reaches the right people. A network that goes quiet after hours is an operational risk, not a technicality.

Designed availability

99.9%

Built on an engineered mast in place of the gum pole it replaces, a Vodacom Business SLA link already confirmed for high availability at this site, and solar/battery power with no single point of failure. CTTX also monitors the network remotely, so an issue is caught before it becomes an outage.

What it's worth over time

Over the 36-month agreement the lodge spends about the same as it would renting — and at the end of it, owns the network. Every month after that is money kept.

Current monthly bill	New bill, once built	Saved over 5 years	Build paid back in
R18,106 /mo	R8,949 /mo	R352,621	2 yr, 10 mo



■ Continue with current provider (cumulative cost)
 ■ Build with CTTX (cumulative cost)

How to read this: the CTTX line starts at the R358,599 paid up front (the R355,000 build plus the R3,599 connection fee). From month 1 both paths pay their monthly bills — R18,106 to the current provider, R8,949 on the CTTX path. The two lines cross at 2 years 10 months. By the end of the 36-month agreement the lodge has paid R680,763 on the CTTX path against R705,336 to continue renting — R24,573 less — and owns the mast, radios, Wi-Fi and power system outright. Over five years the difference is R352,621.

Assumptions, stated plainly: CTTX's monthly fee is fixed for the 36-month term and assumed to rise 8% a year after it. The current provider's bill is assumed to rise 8% a year from year two (a typical CPI-linked telecom escalation; not confirmed by them). With no escalation on either side, payback is 3 years 4 months. The full year-by-year workings are available as a spreadsheet on request.

What the chart doesn't show

Three things the monthly comparison understates — all of them in the lodge's favour.

AN ASSET, NOT AN EXPENSE

The build is capital expenditure: R308,696 excluding VAT goes onto the balance sheet as owned infrastructure — mast, power system, radios, Wi-Fi — rather than leaving as rent every month. For a VAT-registered entity the R46,304 input VAT is recoverable, and the asset can be depreciated over its working life. It adds to what the property is worth and transfers with it. The lodge's accountant will confirm the exact treatment.

GROWING COSTS ENDPOINTS, NOT FEES

The red line is today's bill — it does not yet include the spa, because the spa is not connected. On the current model, reaching it means a wireless link plus more rented access points, each at roughly R263 a month for as long as the lodge stands (its own invoice rate). The same applies to every camera or new building later. On an owned network, adding a point is a once-off piece of hardware on a mast that already covers it.

A LOCKED PRICE, THEN NO RENT AT ALL

For the 36-month term the connectivity fee and CTTX's support retainer are fixed. When the term ends, the lodge keeps the network and simply renews the carrier link — there is no equipment rental to renew, ever. The current model has no such end: the rental line item continues, and rises, for as long as the service does.

The investment

One figure covers the complete build — design, hardware, mast, power system, installation and commissioning. Paid in two parts, owned outright from day one.

Total build investment	Payment terms	Ownership
R355,000	50% on order 50% on commissioning	100% the lodge's
incl. VAT (R308,696 ex VAT)	the deposit secures the equipment allocation	capitalised as a fixed asset on handover

WIRELESS NETWORK

ePMP 4500L base station radios	×2
Sector antennas, 90/120°	×2
ePMP Force 4525L subscriber radios	×28
Wi-Fi 6 indoor access points	×28
Wi-Fi 6 outdoor access points	×2
Gigabit PoE+ managed switches	×4

POWER SYSTEM

Weatherproof equipment cabinet	×1
MPPT solar charge controller	×1
Lithium battery bank & switchgear	×1
680W solar panels	×4
Grid/solar inverter	×1
Solar combiner & isolators	set

MAST & FOUNDATION

9m galvanized steel lattice mast	×1
Concrete, sand & grit for the plinth	as req.
Reinforcing steel (rebar)	as req.
Foundation shuttering	as req.
Holding-down bolts	set
Mounting brackets & clamps	set

CABLING, EARTHING & INSTALL

Outdoor Ethernet & feeder cable	as req.
Armored power & solar DC cable	as req.
Copper earthing rods & bonding	set
Site survey & RF design	incl.
Installation, 2-technician team	incl.
Commissioning, training & handover	incl.

WHY THIS PRICE IS POSSIBLE

The mast and the equipment cabinet in this build are refurbished, CTTX-built units — engineered infrastructure we have on hand, offered here at a substantially reduced cost against buying new. Bought new, those two items alone would cost close to double. That is a meaningful part of how the total above is what it is, and it applies to this proposal specifically.

Next steps

1 Confirm the design

Line-of-sight for all 28 connection points is already confirmed by field survey. A final walk-through validates cable runs and exact mounting positions.

2 Approve the investment

Sign off on the build and the 36-month connectivity agreement, and pay the 50% deposit — that places the equipment order and secures the allocation. The balance is due on commissioning. The Vodacom feasibility submission goes in at the same time, so it runs in parallel with the build.

3 Build the private network

Mast, cabinet, power system, radios and access points go in first. This is the part CTTX builds and owns end to end, and it does not wait on Vodacom.

4 Vodacom site survey

Once the mast is standing, CTTX confirms the site is ready and Vodacom surveys against it. Their installation depends on being satisfied with that mast — which is why it has to be built first.

5 Carrier link installed

The last piece: the 100 Mbps symmetrical, uncapped, unshaped, unthrottled SLA link from Vodacom's Olifants Nek high site, installed in one go.

6 Go live

The current connection is decommissioned once the new network is tested and confirmed — including the spa, connected for the first time.

A NOTE ON TIMING

The wireless equipment for this build is in stock with our supplier today, and we have secured it. This kind of kit tightens up at this time of year, and we can't hold an allocation indefinitely — if it is sold on and the build stands over to next year, it will cost more. That is not a reason to decide in a hurry; it is simply the honest position on where the stock sits today.

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This proposal is valid for 30 days from the date above.